

Robust and nonlinear control EEN050

Assignment 4

Linear Parameter Varying Models and Analysis

Before submission pre-approval of solution is mandatory
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Problem formulation and aim

Consider a SISO LPV system for a virus epidemiology model. Observability and quadratic parameter independent analysis will be performed. Some time domain analysis will be conducted. We recommend to use YALMIP for stability analysis (<https://yalmip.github.io/>) due to its transparent coding structure, see more details <https://yalmip.github.io/tutorial/semidefiniteprogramming/> (commands suggested: `sdpvar`, `optimize`, `check`, `value`). Recommended solver is LMILAB available with MATLAB distribution (SEDUMI is an alternative).

Administration

- The assignment is solved in a group of three students, documented in an electronic solution report (plots, explanation, reply to questions).
- Pre-approval of solution by TAs is mandatory before submission (during tutorial sessions)
- Electronic submission to CANVAS, upload 1 solution per group using the filename: Group#-Assignment#.pdf after getting a preliminary oral approval from the TA (tutorial session).
- Results, no later than 1 week after submission (with constructive feedback). If the solution is not approved, one week time is given for correction (one extra chance only).
- Deadline: check the course PM.

Questions

Given a control oriented modified SEIR (Susceptible-Exposed-Infected-Recovered) epidemiological compartmental model [8,7] as

$$\begin{aligned}\dot{x}_1(t) &= -u(t)x_1(t)x_3(t) + c_1x_4(t) \\ \dot{x}_2(t) &= u(t)x_1(t)x_3(t) - c_2x_2(t) \\ \dot{x}_3(t) &= -c_3x_3(t) + c_2x_2(t) \\ \dot{x}_4(t) &= c_3x_3(t) - c_1x_4(t)\end{aligned}$$

where $x \in \mathbb{R}^4$ x_1 , x_2 , x_3 and x_4 are the susceptible, exposed, infected and recovered fraction of the population. Parameters are given as $c_1 = 1/60, c_2 = 0.2, c_3 = 0.1$ (coefficients with unit $1/day$). $u(t)$ is kept *constant* control input describing the effect of quarantining/self isolation/social distancing in unit $1/(people \cdot day)$, $0 \leq u(t) \leq 1$. The traditional SEIR model is expanded; recovered people can be re-infected.

Exercise 1 Transform the SEIR model into Linear Parameter Varying form under constant control input actions $\forall t$. Use a minimum number of time varying parameters $p(t)$ while creating an LPV model. Classify the obtained model (exo or endo, linear, polynomial or rational in the parameter, exact or non-exact). Simulate the LPV model with constant control inputs $u(t) = 0$ (complete lockdown), $u(t) = 0.3$ (mild distancing), $u(t) = 0.8$ (no distancing). Select a proper initial condition, $x_0 = x(0)$ (we describe a population via the elements of x) and run a time domain situation. How does $x(t)$ behave in time domain $t \in [0 \ T]$ with a preselected terminal time started from $x(0)$? What is the relation between $\sum_{i=1}^4 x_i(T)$ and $\sum_{i=1}^4 x_i(0)$ and why?

Exercise 2 Given we measure the infected amount of people, is the LPV model structurally observable? Find the condition that helps to decide such property.

Exercise 3 Define a range for the parameter ($0 \leq p(t) \leq 0.6$) selected in Ex 1. Is the system model quadratically stable with parameter independent Lyapunov matrix? Use the constant $c = 0.1$. Note: Verify any solutions returned by the solver (see e.g., the `check` command).

Regardless of the LMI result, is the LPV model asymptotically stable? Justify your answer, for example by relating it to your findings in Ex 1.

Exercise 4 Use AI to answer this question, but critically evaluate the answer (explain and interpret) to TAs. It is difficult to derive Lyapunov-based (L)MI stability tests (both necessary and sufficient conditions) for quasi-LPV models (with a connection between parameters and states). Why is this the case? Suggest two meaningful and concise research directions as solutions to this problem for potential master's theses.

Before submission pre-approval of solution is mandatory

References

- [1] S. Skogestad, I. Postletwaite *Multivariable feedback control: analysis and design*. Wiley, New York, 2nd edition, 2001

- [2] HK Khalil *Nonlinear Systems*. Prentice Hall, Upper Saddle River, NJ, 1996
- [3] Jeff Moehlis https://sites.me.ucsb.edu/~moehlis/APC514/tutorials/tutorial_seasonal/node4.html.
- [4] G Horvath, G Szederkenyi *Identifiability analysis and parameter estimation of a SEIR epidemic model*. In Proc of PCIT, 2020 <https://pcit2020.mik.uni-pannon.hu/images/program/kiadvany.pdf>.